

TSUNAMI

DEFINITION:

is a Japanese word meaning “harbour wave”. These waves, which often affect distant shores, originate from undersea or coastal seismic activity, landslides, and volcanic eruptions. Whatever the cause, sea water is displaced with a violent motion and swells up, ultimately surging over land with great destructive power.



ONSET TYPE and CAUSES

If the earthquake or under water land movement is near the coast then tsunami may strike suddenly and if the earth movement is far in the sea then it may take few minutes to hours before striking the coast. The onset is extensive and often very destructive. The general causes of Tsunamis are geological movements. It is produced in *three major ways*. The most common of these is fault movement on the sea floor, accompanied by an earthquake. To say that an **earthquake** causes a tsunami is not completely correct. Rather, both earthquakes and tsunamis result from fault movements. Probably the second most common cause of tsunamis is a **landslide** either occurring underwater or originating above the sea and then plunging into the water. The third major cause of tsunamis is **volcanic activity**. The flank of a volcano, located near the shore or underwater, may be uplifted or depressed similar to the action of a fault. Or, the volcano may actually explode. In 1883, the violent explosion of the famous volcano, Krakatoa in Indonesia, produced tsunamis measuring 40 meters which crashed upon Java and Sumatra. over 36,000 people lost their lives as a result of tsunami waves from Krakatoa. The giant tsunamis that are capable of crossing oceans are nearly always

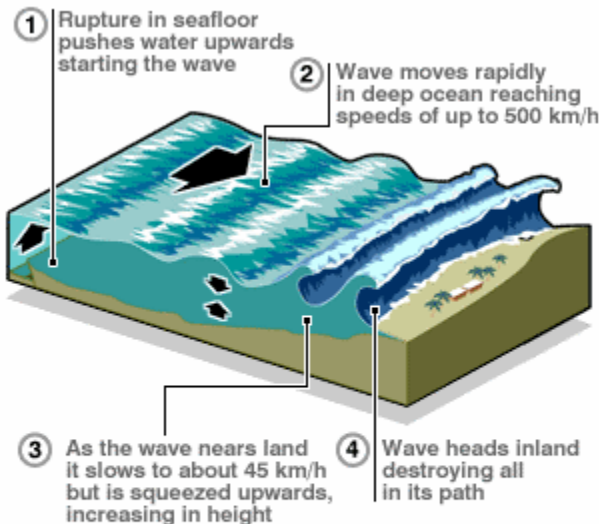
26th December, 2004, A massive earthquake of Magnitude 9.0 hit Indonesia generating Tsunami waves in South-east Asia & eastern coast of India. Height of tsunami waves ranged from 3 – 10 m affecting a total coastal length of 2260 km in the States of Andhra Pradesh, Tamil Nadu, Kerala & UTs of Pondicherry, Andaman & Nicobar Islands. Tsunami waves travelled upto a depth of 3 km from the coast killing more than 10,000 people & affected more than lakh of houses leaving behind a huge trail of destruction.

" t o w a r d s s a f e r I N D I A "

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created by movement of the sea floor associated with earthquakes which occur beneath the sea floor or near the

- ① Rupture in seafloor pushes water upwards starting the wave
- ② Wave moves rapidly



Tsunami wave train formation: Seen in the figure is the rupture in the seafloor shunted in the vertical direction. This movement displaces hundreds of cubic kilometres of the overlaying water, generating a massive tsunami, or sea surge.

How is a Tsunami formed?

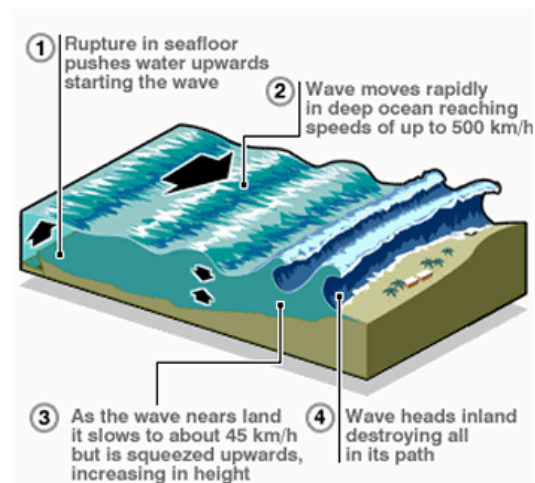
- Seismicity generated tsunamis are result of abrupt deformation of sea floor resulting vertical displacement of the overlying water.
- When earthquakes occur beneath the sea level, the water above the reformed area is displaced from its equilibrium position.
- The release of energy produces tsunami waves which have small amplitude but a very long wavelength.
- It may be caused by non-seismic event also such as a landslide or impact of a meteor.

created by movement of the sea floor associated with earthquakes which occur beneath the sea floor or near the ocean.

WARNING

Tsunami is not a single giant wave. It consists of ten or more waves which is termed as a "tsunami wave train". Since scientists cannot predict when earthquakes will occur, they cannot predict exactly when a tsunami will be generated. Studies of past historical tsunamis indicate where tsunamis are most likely to be generated, their potential heights, and flooding limits at specific coastal locations. With use of satellite technology it is possible to provide nearly immediate warnings of potentially tsunamigenic earthquakes. Warning time depends upon the distance of the epicenter from the coast line. The warning includes predicted times at selected coastal communities where the tsunami could travel in a few hours. In case of tsunamigenic earthquakes or any other geological activity people near to the coastal areas may get very little time to evacuate on receiving of warning.

Note: Ground shaking signals the occurrence of the earthquake. It is important to move away from the coastal areas since a tsunami may accompany the earthquake.



Tsunami wave train formation: Seen in the figure is the rupture in the seafloor shunted in the vertical direction. This movement displaces hundreds of cubic kilometres of the overlying water, generating a massive tsunami, or sea surge.

Early warning systems – a well networked system in place can warn the communities of the coastal areas when the threat is perceived. Tsunami warning should be disseminated to local, state, national as well as the international community so as to be prepared as they are capable of crossing across continents. The information can be broadcasted to the local emergency officers and the general public. On receiving of the warning the action should be to evacuate the place as decided in the evacuation plan.

Tsunami Warning and Communication System

- The Indian Tsunami Early Warning Centre (ITEWC) has been established at Indian National Centre for Ocean Information Sciences, (INCOIS - ESSO) Hyderabad.
- It has the responsibility to provide tsunami advisories to Indian Mainland and the Island regions

In 1995 the US National Oceanic and Atmospheric Administration (NOAA) began developing the Deep Ocean Assessment and Reporting of Tsunami (DART) system. By 2001 six stations had been deployed in the Pacific Ocean. Each station consists of a sea bed bottom pressure recorder (at a depth of about 6000 m) which detects the passage of a tsunami and transmits the data to a surface buoy. The surface buoy then radios the information to the PTWC.

International Tsunami Warning Systems: Shortly after the Hilo Tsunami (1946), the Pacific Tsunami Warning System (PTWS) was developed with its operational center at the Pacific Tsunami Warning Center (PTWC) near Honolulu, Hawaii. The PTWC is able to alert countries several hours before the tsunami strikes. The warning includes predicted arrival time at selected coastal communities where the tsunami could travel in few hours. A tsunami watch is issued with subsequent arrival time to other geographic areas.

Regional Warning Systems usually use seismic data about nearby earthquakes to determine if there is a possible local threat of a tsunami. Such systems are capable enough to provide warnings to the general public in less than 15 minutes.

In India, the *Survey of India* maintains a tide gauge network along the coast of India. The gauges are located in major ports as shown in the figure 2.2.4. The day-to-day maintenance of the gauge is carried with the assistance from authorities of the ports.

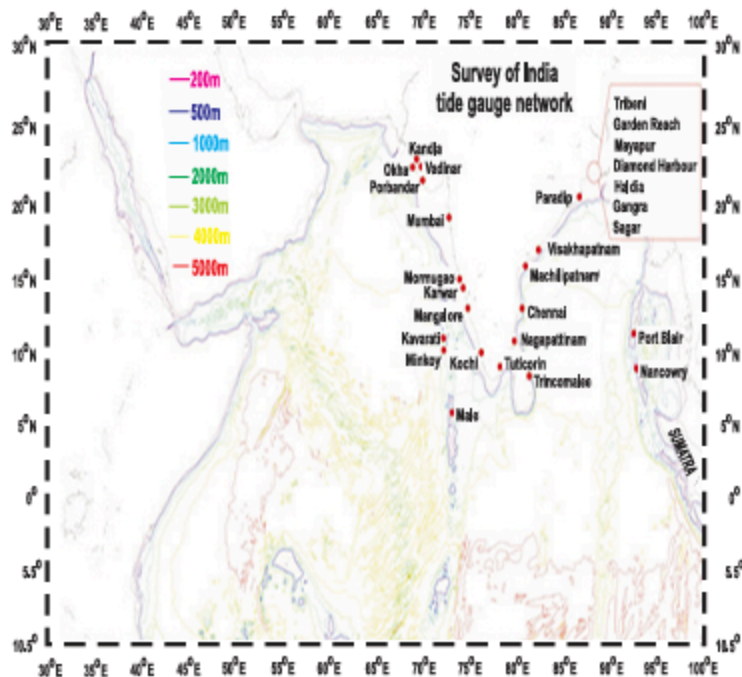


Fig. 2.2.4 : Tide gauge network in India

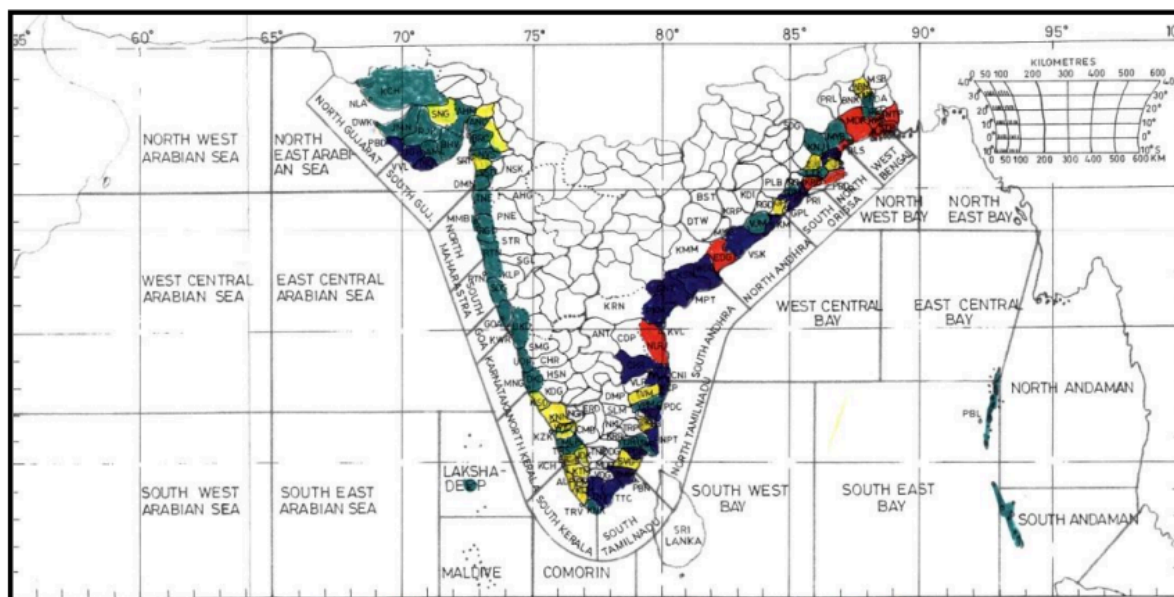
Apart from the tide gauge, tsunami can be detected with the help of radars. The 2004 Indian Ocean tsunami, recorded data from four radars and recorded the height of tsunami waves two hours after the earthquake. It should be noted that the satellites observations of the Indian Ocean tsunami would not have been of any use in

delivering warnings, as the data took five hours to process and it was pure chance that the satellites were overhead at that time. However, in future it is possible that the space-based observation might play a direct role in tsunami warning.

ELEMENTS AT RISK

All structures located within 200 m of the low lying coastal area are most vulnerable to the direct impact of the tsunami waves as well as the impact of debris & boulders brought by it. Settlements in adjacent areas will be vulnerable to floods & scour. Structures constructed of wood, mud, thatch, sheets and structures without proper anchorage to foundations are liable to be damaged by tsunami waves & flooding. Other elements at risk are infrastructure facilities like ports & harbours, telephone and electricity poles, cables. Ships & fishing boats/nets near the coast also add to the destruction caused by tsunami waves.

TSUNAMI - HAZARD ZONATION :



INDIAN CONTEXT :

Distribution pattern of Tsunami in India:

Even though India has not faced frequent Tsunamis but there is a need to identify the areas that are generally affected by Tsunamis. The whole of the Indian coastal belt is prone to Tsunami. Table 2.2.1 shows incidents of tsunamis that have affected our country.

Table 2.2.1: History of tsunami's in India

Date	Location	Impact
1524	Near Dabhol, Maharashtra	Sufficient data not available
02 April 1762	Arakan Coast, Myanmar	Sufficient data not available
16 June 1819	Rann of Kachchh, Gujarat	Sufficient data not available
31 October 1847	Great Nicobar Island	Sufficient data not available
31 December 1881	An earthquake of 7.9 in the Richter scale in Car Nicobar Island	Entire east coast of India and Andaman & Nicobar Islands; 1m tsunamis were recorded at Chennai.
26 August 1883	Explosion of the Krakatoa volcano in Indonesian.	East coast of India was affected; 2m tsunamis were recorded at Chennai.
26 June 1941	An 8.1 Richter scale earthquake in the Andaman archipelago.	East coast of India was affected but no estimates of height of the tsunami is available
27 November 1945	An 8.5 Richter scale earthquake at a distance of about 100km south of Karachi	West coast of India from north to Karwar was affected; 12m tsunami was felt at Kandla.
26 December 2004	Banda Aceh, Indonesia; Tamil Nadu, Kerala, Andhra Pradesh, Andaman and Nicobar Islands, India; Sri Lanka; Thailand; Malaysia; Kenya; Tanzania	The East cost of India was affected. The waves measured around 10 m high killing more than 10,000 precious lives.

TYPICAL EFFECTS

Physical damage - Local tsunami events or those less than 30 minutes from the source cause the majority of damage. The force of can raze everything in its path. It is the flooding effect of a tsunami, however, that most greatly effects human settlements by water damage to homes and businesses, roads, bridges and other infrastructure. Ships, port facilities, boats/trawlers, fishing nets also get damaged.

Environmental damage - There is evidence of ever increasing impact upon the environment on account of the effects of tsunamis. The range varies from generation of tonnes of debris on account of structural collapse of weaker buildings, release of toxic chemicals into the environment on account of chemical leak/spillage/process failure/utility breakages/ collateral hazards and negative impact on the already fragile ecosystems.

Casualties and public health: Deaths occur principally from drowning as water inundates homes or neighborhoods. Many people may be washed out to sea or crushed by the giant waves. There may be some injuries from battering by debris and wounds may become contaminated.

Water supply: sewage pipes may be damaged causing major sewage disposal problems. Drinking water shortage arises due to breakage of water mains and contamination. Open wells and ground water may become unfit for drinking due to contamination of salt water and debris.

Standing Crops and food supplies: flooding by tsunami causes damage to the standing crops and also to the food supplies in the storage facilities. The land may be rendered infertile due to salt water incursion from the sea.



Men walk pass damaged fishing trawlers hit by a tsunami at Nagapattinam port



An aerial view of a bridge destroyed by tsunamis near Pondicherry, in the southern Indian state of Tamil Nadu.



An aerial view of Pattinapakam, a slum which was destroyed by a tsunami, near Madras.

A tsunami can kill or injure people and damage or destroy buildings and infrastructure as waves come in and go out. A tsunami is a series of enormous ocean waves caused by earthquakes, underwater landslides, volcanic eruptions, or asteroids. Tsunamis can:

- Travel 20-30 miles per hour with waves 10-100 feet high.
- Cause flooding and disrupt transportation, power, communications, and the water supply.
- Happen anywhere along U.S. coasts. Coasts that border the Pacific Ocean or Caribbean have the greatest risk.

Most deaths caused by Tsunamis are because of Drowning.

Tsunami – Before, During, After Tsunami & Do's & Don't's

IF YOU ARE UNDER A TSUNAMI WARNING:

- First, protect yourself from an Earthquake. **Drop, Cover, then Hold On.**
- Get to high ground as far inland as possible. You can protect yourself from a tsunami while also protecting yourself and your family from COVID-19. Protect yourself from the effects of a tsunami by moving from the shore to safe, high grounds outside tsunami hazard areas.
- Be alert to signs of a tsunami, such as a sudden rise or draining of ocean waters.
- Listen to emergency information and alerts.
- Evacuate: DO NOT wait! Leave as soon as you see any natural signs of a tsunami or receive an official tsunami warning.
 - Understand that your regular public shelter may not be open this year as shelter locations may have changed due to the COVID-19 pandemic. CDC, FEMA, and the American Red Cross are advising shelter operators on safety procedures to make sure that shelters can protect people from natural disasters while taking COVID-19 precautions. Check with local authorities for the latest information about public shelters or download the free [Red Cross Emergency app](#) for a list of [open Red Cross shelters](#) in your area. In addition:
 - Always follow the instructions from local emergency managers. They provide the latest recommendations based on the threat in your community.
 - Make plans to shelter with friends and family, if possible.

If you are in a room, go out to sea.

HOW TO STAY SAFE WHEN A TSUNAMI THREATENS

Prepare NOW

- If you live near, or regularly visit a coastal area, learn about the risk of tsunami in the area. Some at-risk communities have maps with evacuation zones and routes. If you are a visitor, ask about community plans.
- Learn the signs of a potential tsunami, such as an earthquake, a loud roar from the ocean, or unusual ocean behavior, such as a sudden rise or wall of water or sudden draining of water showing the ocean floor.
- Know and practice community evacuation plans and map out your routes from home, work, and play. Pick shelters 100 feet or more above sea level, or at least one mile inland.
- Create a family emergency communication plan that has an out-of-state contact. Plan where to meet if you get separated.
- Sign up for your community's warning system. The Emergency Alert System (EAS) and National Oceanic and Atmospheric Administration (NOAA) Weather Radio also provide emergency alerts. Sign up for [email updates](#) about COVID-19 from the Centers for Disease Control and Prevention (CDC).
- Consider earthquake insurance and a flood insurance policy through the National Flood Insurance Program (NFIP). Standard homeowner's insurance does not cover flood or earthquake damage.

Survive DURING

- If you are in a tsunami area and there is an earthquake, then first protect yourself from the earthquake. Drop, Cover, and Hold On. Drop to your hands and knees. Cover your head and neck with your arms. Hold on to any sturdy furniture until the shaking stops. Crawl only if you can reach better cover, but do not go through an area with more debris. If possible, avoid touching your eyes, mouth, and nose, especially after touching high-touch surfaces, to slow the spread of COVID-19.
 - When the shaking stops, if there are natural signs or official warnings of a tsunami, then move immediately to a safe place as high and as far inland as possible. Listen to the authorities, but do not wait for tsunami warnings and evacuation orders.
 - If you are outside of the tsunami hazard zone and receive a warning, then stay where you are unless officials tell you otherwise.
 - Leave immediately if you are told to do so. Evacuation routes are often marked by a wave with an arrow in the direction of higher ground.
 - If you are in the water, then grab onto something that floats, such as a raft, tree trunk, or door. There is no evidence that COVID-19 can be spread through water, however floodwaters may contain debris, chemicals, or waste that are harmful to your health.
 - If you are in a boat, then face the direction of the waves and head out to sea. If you are in a harbor, then go inland.
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Be Safe AFTER

- Listen to local alerts and authorities for information on areas to avoid and shelter locations.
- Avoid wading in floodwater, which can contain dangerous debris. Water may be deeper than it appears.
- Be aware of the risk of electrocution. Underground or downed power lines can electrically charge water. Do not touch electrical equipment if it is wet or if you are standing in water.
- If you become injured or sick and need medical attention, contact your healthcare provider for further care instructions and shelter in place, if possible. If you are experiencing a medical emergency, call 9-1-1 and let the operator know if you have, or think you might have, COVID-19. If possible, put on a mask before help arrives. If staying at a shelter or public facility, alert shelter staff immediately so they can call a local hospital or clinic.
- Stay away from damaged buildings, roads, and bridges.
- Document property damage with photographs. Conduct an inventory and contact your insurance company for assistance.
- Save phone calls for emergencies. Phone systems are often down or busy after a disaster. Use text messages or social media to communicate with family and friends.

MITIGATION MEASURES

SPECIFIC PREPAREDNESS MEASURES

Hazard mapping – a hazard map should be prepared with designated areas expected to be damaged by flooding by tsunami waves. Historical data could be of help in showing areas inundated in the past. Keeping in mind the vulnerable areas, evacuation routes should be constructed and mapped. The plan should be followed by evacuation drill.

Early warning systems – a well networked system in place can warn the communities of the coastal areas when the threat is perceived. Tsunami warning should be disseminated to local, state, national as well as the international community so as to be prepared as they are capable of crossing across continents. The information can be broadcasted to the local emergency officers and the general public. On receiving of the warning the action should be to evacuate the place as decided in the evacuation plan.

Community Preparedness – communities in the coastal areas are faced by the wrath of cyclones, storm surge and tsunami waves. It is important that the community is better prepared to take suitable actions on receiving of the threat and follow emergency evacuation plans and procedures. A community which chooses to ignore warning may get severely affected if they are not prepared to take immediate measures.

MAIN MITIGATION STRATEGIES

Site Planning and Land Management- Within the broader framework of a comprehensive plan, site planning determines the location, configuration, and density of development on particular sites and is, therefore, an important tool in reducing tsunami risk.

- The designation and zoning of tsunami hazard areas for such open-space uses as agriculture, parks and recreation, or natural hazard areas is recommended as the first land use planning strategy. This strategy is designed to keep development at a minimum in hazard areas.
- In areas where it is not feasible to restrict land to



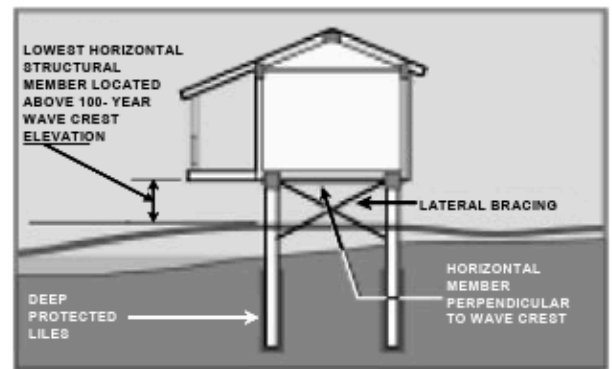
Bad siting - A view of a village that was destroyed by a tsunami at Kolachal, in Tamil Nadu

open-space uses, other land use planning measures can be used. These include strategically controlling the type of development and uses allowed in hazard areas, and avoiding high-value and high-occupancy uses to the greatest degree possible.

- The capital improvement planning and budgeting process can be used to reinforce land use planning policies.

Engineering structures – As most of the structures along the coast area comprises of fisherman community, which are constructed of light weight materials without any engineering inputs. Therefore there is an urgent need to educate the community about the good construction practices that they should adopt such as:

- Site selection – Avoid building or living in buildings within several hundred feet of the coastline as these areas are more likely to experience damage from tsunamis.
- Construct the structure on a higher ground level with respect to mean sea level.
- Elevate coastal homes: Most tsunami waves are less than 3 meters in height. Elevating house will help reduce damage to property from most tsunamis. Structural columns resist the impact while other walls are expendable. It is important to also take note that adequate measures are also brought into the design to cater for earthquake forces.
- Construction of water breakers to reduce the velocity of waves.
- Use of water & corrosion resistant materials for construction.
- Construction of community halls at higher locations, which can act as shelters at the time of a disaster.



Flood management - Flooding will result from a tsunami. Tsunami waves will flood the coastal areas. Flood mitigation measures could be incorporated (see section on floods for additional information). Building barriers or buffers such as special breakwaters or seawalls can be an effective risk reduction measure against gushing waters in case of Tsunami/Storm surge during cyclones.

While it is of course not possible to prevent a tsunami, in certain tsunami prone countries some measures have been taken to reduce the damage caused on shore. Japan has implemented an extensive programme of building *tsunami walls* of up to 4.5m (13.5 ft) high in front of populated coastal areas. Other localities have built *flood gates* and channels to redirect the water from incoming tsunamis. However,



Fig 2.2.5 Tsunami walls in populated coastal areas of Japan



Measures for Safety from Tsunamis and Storm Surges in Coastal areas:

Structural measures:

1. Plantation of mangroves and coastal forests along the coast line
2. Development of a network of local knowledge centers (rural/urban) along the coast lines to provide necessary training and emergency communication during crisis time (e.g. centers developed by M.S. Swaminathan Foundation in Pondicherry)
3. Construction of location specific sea walls and coral reefs in consultation with experts
4. Development of break waters along the coast to provide necessary cushion against tsunami hazards
5. Development of tsunami detection, forecasting and warning dissemination centres
6. Development of a "Bio-Shield" - a narrow strip of land along coastline.
7. Identification of vulnerable structures and appropriate retrofitting for tsunami/cyclone resistance of all such buildings as well as appropriate planning, designing, construction of new facilities like:

- Critical infrastructures e.g. power stations, warehouses, oil and other storage tanks etc. located along the coastline.
- All other infrastructure facilities located in the coastal areas
- Public buildings and private houses
- All marine structures
- Construction and maintenance of national and state highways and other coastal roads

Non-Structural Measures:

1. Strict implementation of the coastal zone regulations (within 500 m of the high tide line with elevation of less than 15 m above.
2. Mapping the coastal area for multiple hazards, vulnerability and risk analysis up to taluka /village level.
3. Capacity building requirements for the local people and the administration for facing the disasters in wake of tsunami and cyclone
4. Developing tools and techniques for risk transfer in highly vulnerable areas
5. Launching a series of public awareness campaign throughout the coastal area
6. Training of local administration in forecasting warning dissemination and evacuation techniques
7. Awareness generation and training among the fishermen, coast guards, officials from Fisheries department and port authorities and local district officials etc., in connection with evacuation and post tsunami storm surge management activities.
8. Studies focusing on the tsunami risk in India may be taken under NCRM project.

COMMUNITY BASED PREPAREDNESS :

Community Preparedness – communities in the coastal areas are faced by the wrath of cyclones, storm surge and tsunami waves. It is important that the community is better prepared to take suitable actions on receiving of the threat and follow emergency evacuation plans and procedures. a community which choose to ignore warning may get severely effected if they are not prepared to take immediate measures.



CASE STUDY :

26th December, 2004, A massive earthquake of Magnitude 9.0 hit Indonesia generating Tsunami waves in South-east Asia & eastern coast of India. Height of tsunami waves ranged from 3 – 10 m affecting a total coastal length of 2260 km in the States of Andhra Pradesh, Tamil Nadu, Kerala & UTs of Pondicherry, Andaman & Nicobar Islands. Tsunami waves travelled upto a depth of 3 km from the coast killing more than 10,000 people & affected more than lakh of houses leaving behind a huge trail of destruction.

2004 TSUNAMI – INDIAN OCEAN

1. https://www.preventionweb.net/files/workspace/7935_casestudy1.pdf

2011 TSUNAMI - JAPAN

2. <https://www.rgs.org/media/rfxobyti/mountainsvolcanoesandearthquakeslesson6japanearthquakecasestudy.pdf>